GradientCIO

TOTAL PLAN ALLOCATION STUDY

Northstar Total Plan Allocation Study

Portfolio construction review · prepared for the Investment Committee

AS OF
2026-06-30

TOTAL NAV
$1,000.0M

PUBLIC-EQUITY WEIGHT
42.0%

POLICY BENCHMARK
Northstar Global Equity Index

01 Executive Decision

Manager and issuer concentration near limits

RECOMMENDATION

We recommend retaining the 42.0% total-fund public-equity allocation while moving 4.0 percentage points of the sleeve from concentrated active mandates to a global passive completion mandate S3 S4 S7. Committee action requested: approve the target manager structure and phased funding, subject to manager diligence and concentration limits S3 S8.

CURRENT EXPECTED RETURN	POLICY EXPECTED RETURN	PORTFOLIO RISK	LIQUIDITY STATUS
42.0%	0.0pp	-0.6pp	2.3% NAV
Policy target 42.0% S3	Total-fund target S3	Trailing 1Y versus benchmark S5	Largest issuer; lagged 13F look-through S7

Recommendation and action requested

We recommend moving 4.0 percentage points of the equity sleeve from concentrated active mandates to a global passive completion mandate. Committee action requested: approve the target lineup and phased funding S3 S4 S8.

DECISION SUMMARY

DECISION ELEMENT	PROPOSED ACTION	EVIDENCE	STATUS
Total equity	Retain 42.0% of total fund	S3	Compliant
Active core	Reduce from 57.1% to 47.6% of equity	S4 S8	Proposed
Passive completion	Increase from 28.6% to 38.1%	S4 S8	Proposed

Conditions and limits

Complete manager comparison and diligence, confirm transition costs, and preserve governed manager and issuer concentration limits S3 S7 S8.

Key judgment — Concentration

Observation: Manager and issuer concentration sit near their limits S1. Why it matters: Moving 4.0pp to a passive completion mandate reduces both. Uncertainty: Transition costs are not yet confirmed. What would change the view: Costs that outweigh the concentration benefit.

Key judgment — Sleeve size

Observation: Public equity is 42.0% of total NAV S2. Why it matters: The 4.0pp shift stays inside the sleeve and leaves the total weight unchanged. Uncertainty: The weight moves with equity markets before funding. What would change the view: A drift of more than 2pp before implementation.

Key judgment — Largest issuer

Observation: The largest single-issuer exposure is 2.3% of NAV S3. Why it matters: It is the position closest to the concentration limit. Uncertainty: Look-through holdings are as of the last 13F filing. What would change the view: A newer filing that changes the top position.

02 Mandate and Policy Basis

Portfolio	**Northstar Total Fund** S2
Mandate	**Global public-equity growth** S8
Benchmark	**Northstar Global Equity Index** S5
Horizon	**10 years** S6
Return basis	**Net** S5
Data scope	**Illustrative sample data** S1 S1

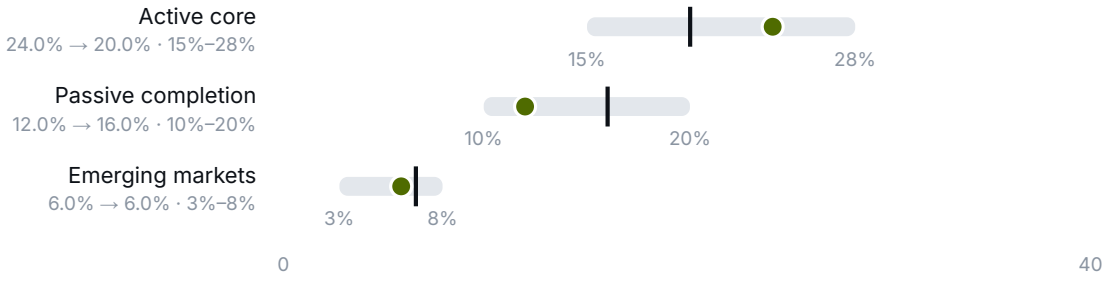
GOVERNED FRAMEWORK

AREA	OBJECTIVE OR LIMIT	OBSERVED	STATUS	SOURCE
Public equity	32.0%–52.0%	42.0%	Compliant	S3
Single manager	At or below 15.0% NAV	13.0%	Near limit	S3 S4
Single issuer look-through	At or below 3.0% NAV	2.3%	Near limit	S3 S7

03 Current, Policy and Candidate Mixes

The proposed mix reduces concentrated risk while preserving mandate exposure

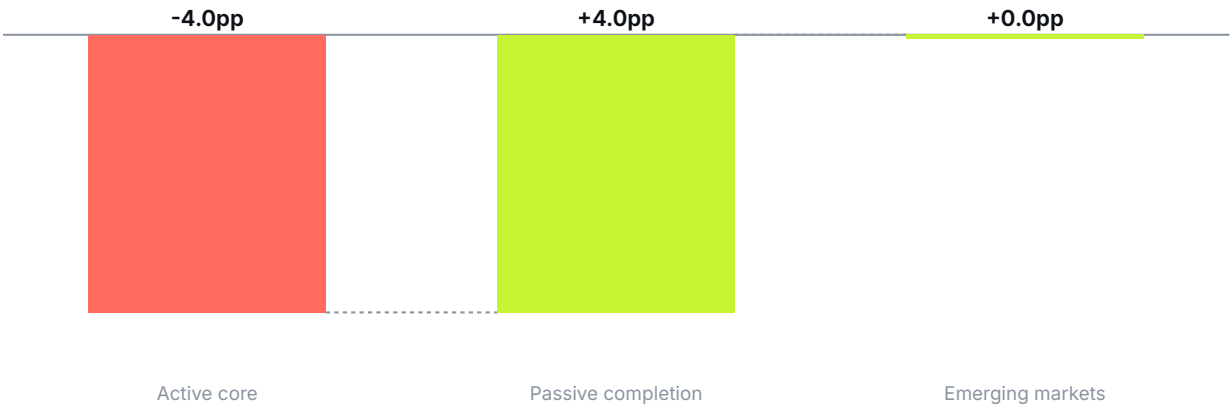
CURRENT WEIGHT VERSUS POLICY RANGE



CURRENT AND TARGET MIX



TARGET ALLOCATION CHANGES (PERCENTAGE POINTS)



CURRENT VERSUS TARGET

SEGMENT	CURRENT	TARGET	CHANGE	RANGE / CONSTRAINT	RATIONALE
Active core	24.0%	20.0%	-4.0pp	15.0%–28.0%	Reduce manager concentration S3 S4
Passive completion	12.0%	16.0%	+4.0pp	10.0%–20.0%	Improve benchmark completion S5 S8
Emerging markets	6.0%	6.0%	0.0pp	3.0%–8.0%	Maintain strategic exposure S3 S6

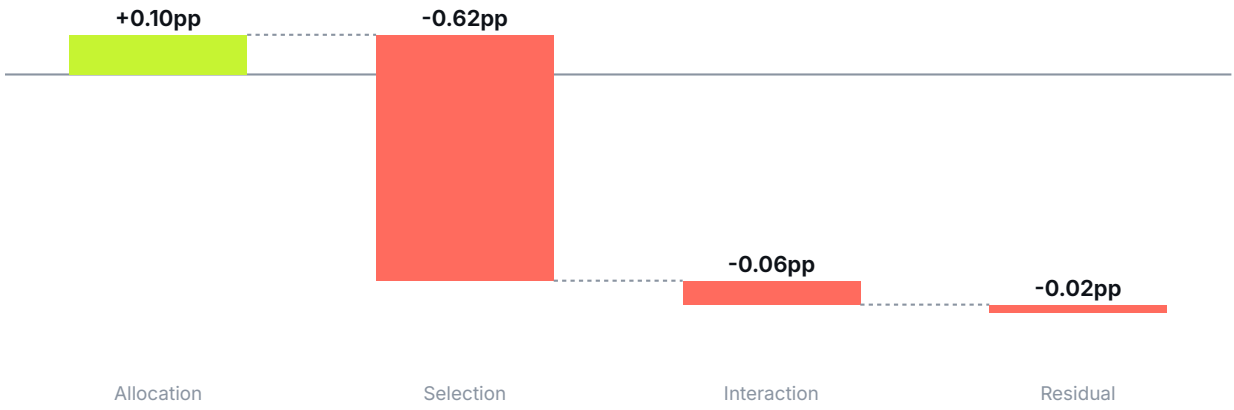
> **Analysis — Why this target**
>
> Observation: The target moves 4.0 percentage points from active core to passive completion S4 S8. Why it matters: Manager concentration falls while equity exposure remains preserved S3. Uncertainty: CMA, regime, findings and cost coverage remain incomplete S9. What would change the view: Completed records showing no measurable concentration reduction S8 S9.

The target is a committee proposal and is not a saved portfolio change S3 S8.

04 Expected Return Assumptions

Separate user and CMA assumptions before comparing candidate mixes

GOVERNED ATTRIBUTION EFFECTS



Returned attribution effects; residual is shown explicitly S5.

STANDARD PERIODS

PERIOD	PORTFOLIO / SLEEVE	BENCHMARK	EXCESS	COVERAGE
1Y equity sleeve	10.2%	10.8%	-0.6pp	Complete S5
3Y equity sleeve (ann.)	9.4%	9.1%	+0.3pp	Complete S5
5Y equity sleeve (ann.)	8.1%	8.4%	-0.3pp	Complete S5

ATTRIBUTION EFFECTS

EFFECT	CONTRIBUTION	SOURCE
Allocation	+0.10pp	S5
Selection	-0.62pp	S5
Interaction	-0.06pp	S5
Residual	-0.02pp	S5

Governed attribution method and reconciliation metadata follow S5.

Method	**Brinson-Fachler** S5
Linking	**Symmetric Carino** S5
Basis	**Net, monthly linked** S5
Tolerance	**1 bp; degraded residual** S5

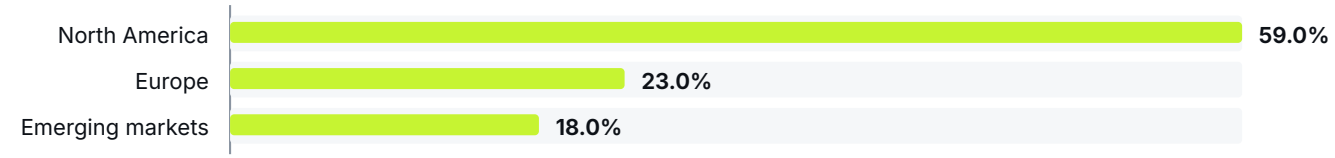
Analysis — Expected-return basis

Observation: User and CMA expected returns are displayed on separate policy-target bases S3 S9. Why it matters: A 40 bp gap can change the allocation decision without changing weights S3 S9. Uncertainty: CMA estimates remain model-dependent and do not replace user assumptions S9. What would change the view: Updated assumptions that reduce the sleeve-level gap below 10 bp S3 S9.

05 CMA Risk and Euler Contribution

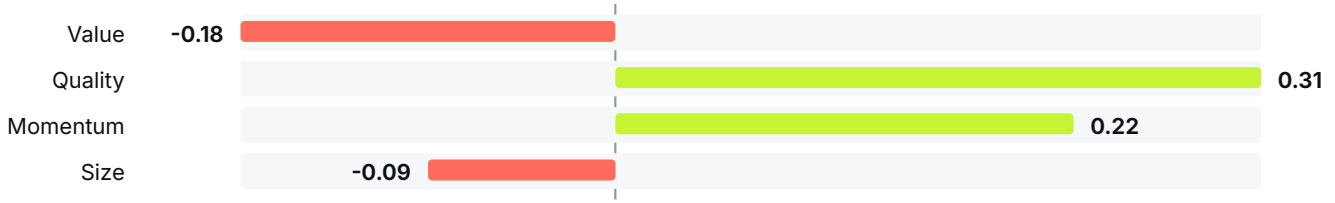
Completion exposure reduces reliance on concentrated active risks

REGIONAL ALLOCATION



Returned equity-sleeve regional allocation S4.

ACTIVE FACTOR EXPOSURE



GOVERNED CONCENTRATION

AXIS	OBSERVED	LIMIT	STATUS	SOURCE
Largest manager	13.0% NAV	15.0%	Near limit	S3 S4
Largest issuer look-through	2.3% NAV	3.0%	Near limit	S3 S7
North America	59.0% of equity	Not assessed	Not assessed	S4

Analysis — Concentration remains near limits

Observation: The largest manager and issuer look-through remain below but near governed limits S3 S4 S7. Why it matters: Completion exposure can reduce reliance on concentrated active mandates. Uncertainty: Issuer look-through evidence is lagged and incomplete S7. What would change the view: Refreshed holdings showing concentration safely below policy thresholds S4 S7.

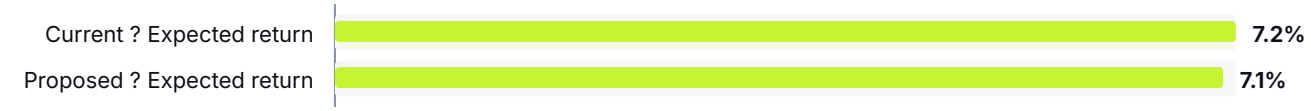
Look-through limitation

The issuer concentration uses lagged Form 13F evidence and does not represent a complete holdings record S7.

06 Regime and Stress Scenarios

Governed assumptions frame tradeoffs rather than forecasts

FORWARD EXPECTED-RETURN COMPARISON



Governed forward assumptions; assumptions, not forecasts S6.

Assumption set	**2026 Strategic CMA** S6
Regime	**Baseline** S6
Horizon	**10 years** S6
Currency	**USD** S6
Context fingerprint	**test-equity-forward-2026** S6

GLOBAL EQUITY FORWARD ASSUMPTIONS

MEASURE	CURRENT SLEEVE	PROPOSED SLEEVE
Expected return	7.2%	7.1%
Expected volatility	16.8%	15.7%
Expected max drawdown	-28.5%	-26.6%

Analysis — Risk falls modestly

Observation: The proposed sleeve has lower expected return, volatility and drawdown than the current sleeve S6. Why it matters: Completion trades little assumed return for lower modeled risk. Uncertainty: These long-horizon assumptions exclude realized implementation results. What would change the view: Refreshed results showing no measurable modeled-risk improvement S6.

These figures are assumptions under the selected context, not forecasts or guarantees S6.

07 Liquidity and Capital-Call Coverage

Concentration limits shape manager implementation

LARGEST GOVERNED CONCENTRATIONS



Returned concentrations and governed limits S3 S4 S7.

CURRENT AND PROPOSED ROLES

MANAGER	ROLE	EXPOSURE	EVIDENCE STATUS	PROPOSED ACTION	NEXT STEP
Example Active Global Manager A	Active core	15.0% NAV	Current S8	Reduce by 2.5pp	Review funding schedule
Example Active Global Manager B	Active core	9.0% NAV	Current S8	Reduce by 1.5pp	Complete performance review
Example Global Index Fund	Passive completion	12.0% NAV	Current S8	Increase by 4.0pp	Confirm vehicle terms

LOOK-THROUGH FLAGS

ISSUER	LOOK-THROUGH VALUE	% NAV	MANAGERS	EVIDENCE DATE	FLAG
Example Technology Issuer A	$23.0M	2.3%	2	2026-05-15	Near policy watch level S7
Example Consumer Issuer B	$15.0M	1.5%	2	2026-05-15	Monitor overlap S7

Form 13F caveat

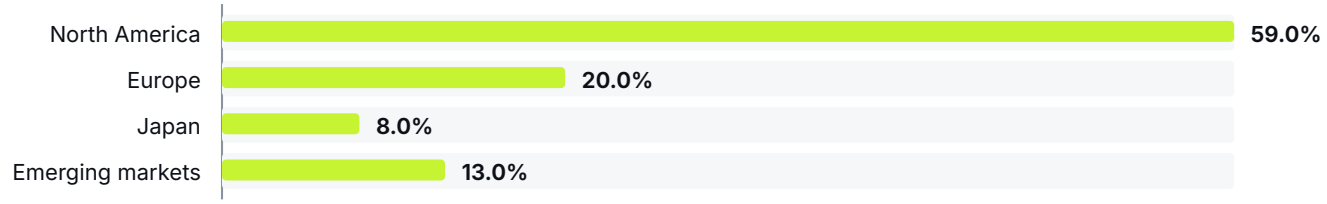
Form 13F evidence is lagged by up to 45 days, covers long-only US-listed equity, excludes shorts, cash, non-US listings and private assets, and uses USD filing values that are not FX-converted against NAV S7.

Analysis — Concentration shapes implementation

Observation: The largest manager and issuer remain below but near governed limits S3 S4 S7. Why it matters: Completion reduces reliance on concentrated active mandates. Uncertainty: Issuer look-through evidence is lagged and incomplete S7. What would change the view: Refreshed manager records showing concentration safely below policy thresholds S4 S7.

08 Commitments and Pacing

PUBLIC-EQUITY REGIONAL EXPOSURE



CURRENT SLEEVE

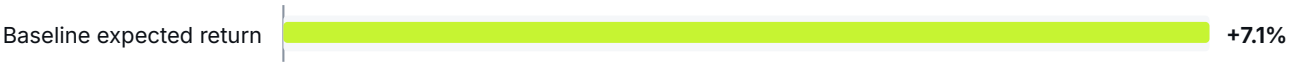
MANAGER / EXPOSURE	ROLE	VALUE	% PORTFOLIO	% EQUITY	AS OF
Example Active Global Manager A	Active core	$150.0M	15.0%	35.7%	2026-06-30 S4
Example Active Global Manager B	Active core	$90.0M	9.0%	21.4%	2026-06-30 S4
Example Global Index Fund	Passive completion	$120.0M	12.0%	28.6%	2026-06-30 S4
Example Emerging Manager	Emerging markets	$60.0M	6.0%	14.3%	2026-06-30 S4

Manager roles are from the illustrative mandate; regional weights are returned portfolio analytics and retain their stated basis S4 S8.

> **Analysis — Pacing evidence gap**
> Observation: Unfunded commitments are reported, but commitment cash flows are not loaded S8. Why it matters: The study cannot infer drawdown timing from NAV or unfunded amounts. Uncertainty: Pacing remains unavailable. What would change the view: Loaded cash flows supporting a governed commitment projection would enable pacing analysis S8.

09 Implementation Constraints

GOVERNED SCENARIO RESULTS



Alternative-regime and tracking-error scenarios are unavailable S9.

SUPPORTED SCENARIOS

SCENARIO	RETURN / LOSS	RISK MEASURE	BASIS	SOURCE
Baseline assumptions	7.1% expected return	15.7% volatility	Proposed sleeve	S6
Alternative regime	Not available — no comparable returned context	Not available	Unavailable	S9
Tracking-error scenario	Not available — active-risk result not returned	Not available	Unavailable	S9

> **Selected-series sandbox — not saved-portfolio analytics**
> Not available — no Strategy Lab return series selected S9.

10 Risks, Open Items and Approvals

IMPLEMENTATION STEPS

PHASE	ACTION	SIZE	FUNDING SOURCE	CONDITION	OWNER
1	Fund passive completion	2.0pp	Active-manager redemption	Vehicle and liquidity confirmed	Investment staff
2	Complete target shift	2.0pp	Active-manager redemption	Diligence and transition costs complete	CIO

ALTERNATIVES CONSIDERED

ALTERNATIVE	BENEFIT	COST / RISK	REASON NOT SELECTED
No change	No turnover	Concentration persists	Does not address watch items S3 S7
Replace all active core	Maximum simplification	Manager and transition risk	Evidence does not support full replacement S5 S8

- Complete manager comparison and diligence before final funding S8.
- Confirm transition costs before implementation S9.
- Refresh governed concentration and attribution after phase one S3 S5.

11 Coverage

Portfolio and allocation	**Available**	Illustrative sample data S2
Policy	**Available**	Allocation and concentration S3
Exposure	**Available**	Manager and regional rows S4
Returns and attribution	**Incomplete**	Residual outside tolerance S5
Forward assumptions	**Available**	2026 Strategic CMA S6
13F look-through	**Incomplete**	Lagged and incomplete S7
Manager evidence	**Available**	illustrative mandate S8
Scenario and cost evidence	**Not available**	Not returned S9
CMA consensus	**Not available**	Optional context not returned S6
Strategy Lab	**Not available**	No selected series S9
Manager diligence	**Not available**	Optional diligence not returned S8
Transaction costs	**Not available**	Not returned S9

A Appendix — Sources and methods

Based on 9 of 10 evidence areas.

TAG	EVIDENCE	SOURCE	AS OF	CHECK
S1	Organization and capabilities	list_organizations; get_gradient_capabilities	2026-06-30	**Passed**
S2	Portfolio and allocation	list_portfolios; get_portfolio_structure	2026-06-30	**Passed**
S3	Policy assessment	check_portfolio_policy	2026-06-30	**Passed**
S4	Public-equity exposure	get_portfolio_exposure	2026-06-30	**Passed**
S5	Returns and attribution	get_portfolio_historical_returns; get_portfolio_attribution	2026-06-30	**Incomplete**
S6	Forward assumptions	get_capital_market_assumptions	2026-06-15	**Passed**
S7	13F look-through	get_portfolio_exposure	2026-05-15	**Advisory**
S8	Manager mandate evidence	get_gradient_capabilities	2026-06-30	**Passed**
S9	Scenario and cost coverage	get_gradient_capabilities	2026-06-30	**Advisory**

Illustrative, Gradient Maintained — demo data, not the client's holdings or managers. All organizations, portfolios, managers and results in this public example are illustrative.

SOURCE AUDIT TRAIL

TAG	KEY PARAMETERS	DIGEST
S1	organization=test	sha256:aaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaa
S2	allocation_tree	sha256:aaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaa
S3	portfolio=test	sha256:aaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaa
S4	public_equity; allocations	sha256:aaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaa
S5	all sections	sha256:aaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaa
S6	baseline	sha256:aaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaa
S7	include=lookthrough_13f	sha256:aaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaa
S8	illustrative	sha256:aaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaa
S9	illustrative	sha256:aaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaa

Data scope for every source: Illustrative sample data. Each digest is the SHA-256 fingerprint of the exact data the report used.

B Appendix — Server Metric Methods and Disclosures

SERVER METRIC METHODS

SOURCE	QUANTITY	METHOD / FORMULA	VERSION	BASIS / UNITS / TOLERANCE
S3	Policy status	Governed assessment	portfolio-policy-v1	Returned status
S5	Returns and attribution	Linked returns; Brinson-Fachler	portfolio-attribution-v1	Net; 1 bp tolerance

- Forward values are assumptions, not forecasts S6; Form 13F limits are set out in the caveat above S7.
- Strategy Lab results are calculated separately. AI-assisted; not investment, legal or compliance advice.